

Overview of Large Language Models

Contents

01

History of Large Language Models (LLMs)

현재 AI 업계의 핵심인 Large Language Models (LLMs)이 무엇인지 명확히 이해하고, 이에 기본이 되는 개념들을 학습합니다.

02

How to improve LLMs?

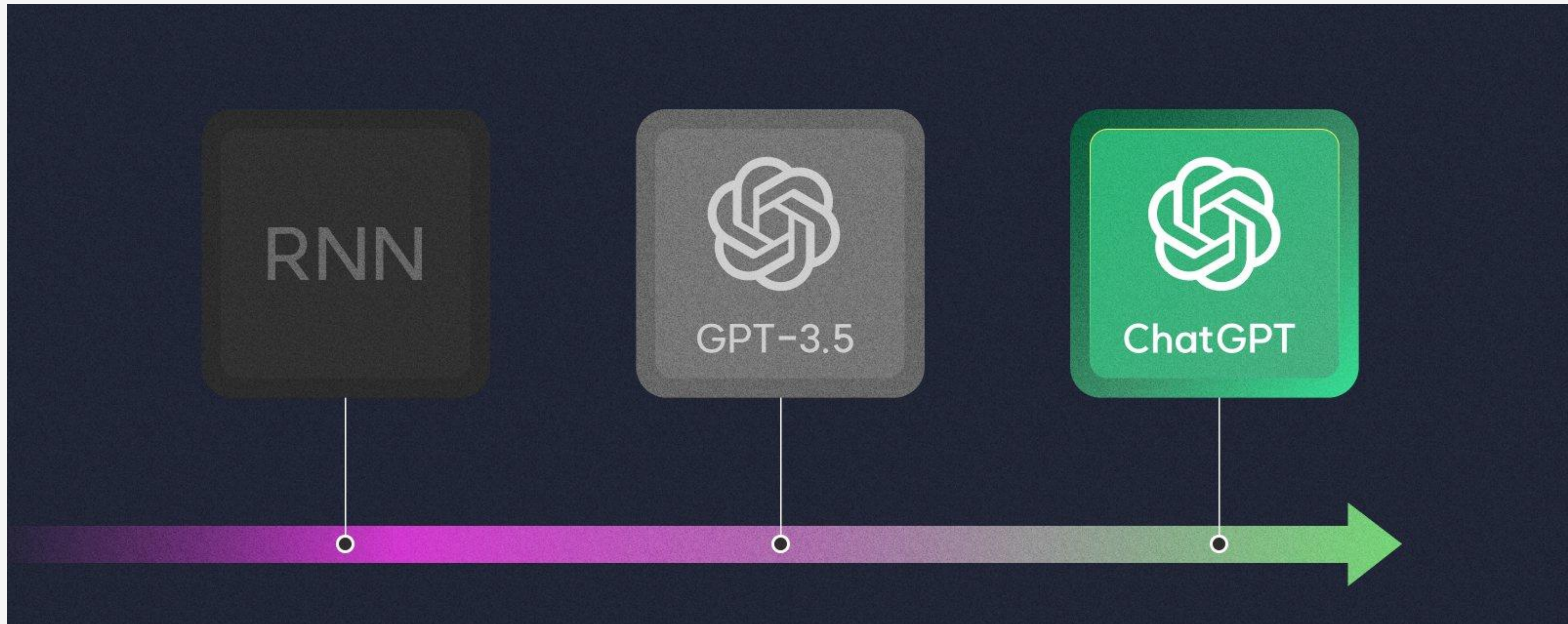
LLMs의 성능을 높이기 위한 다양한 techniques (prompt engineering, RAG, instruction tuning, alignment tuning)에 대해 이해합니다.

History of Large Language Models

(GPT series를 중심으로)

ChatGPT of OpenAI

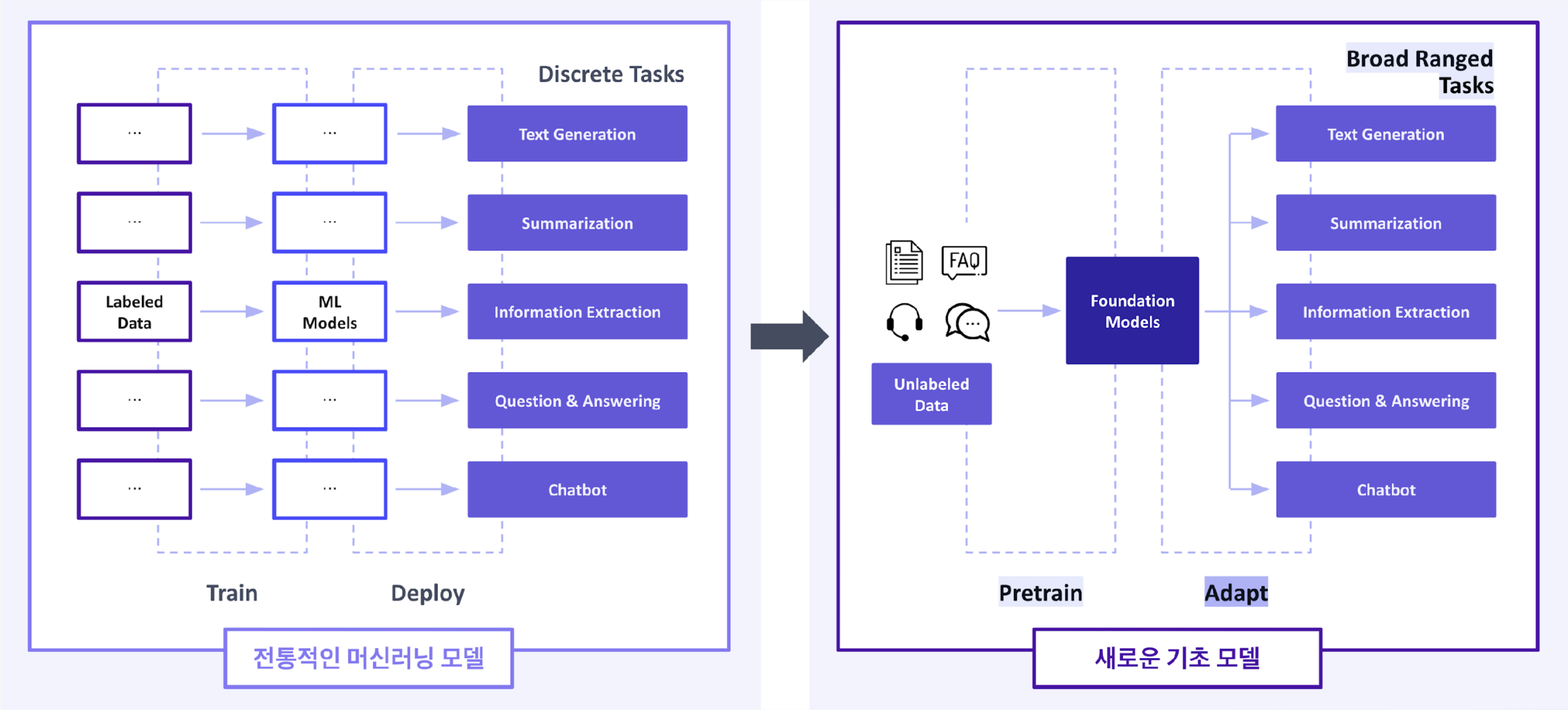
[1]



[1] <https://www.upstage.ai/blog/insight/evolution-of-gpt-models>

Reset Moment : 거대한 전환의 신호탄

[2]

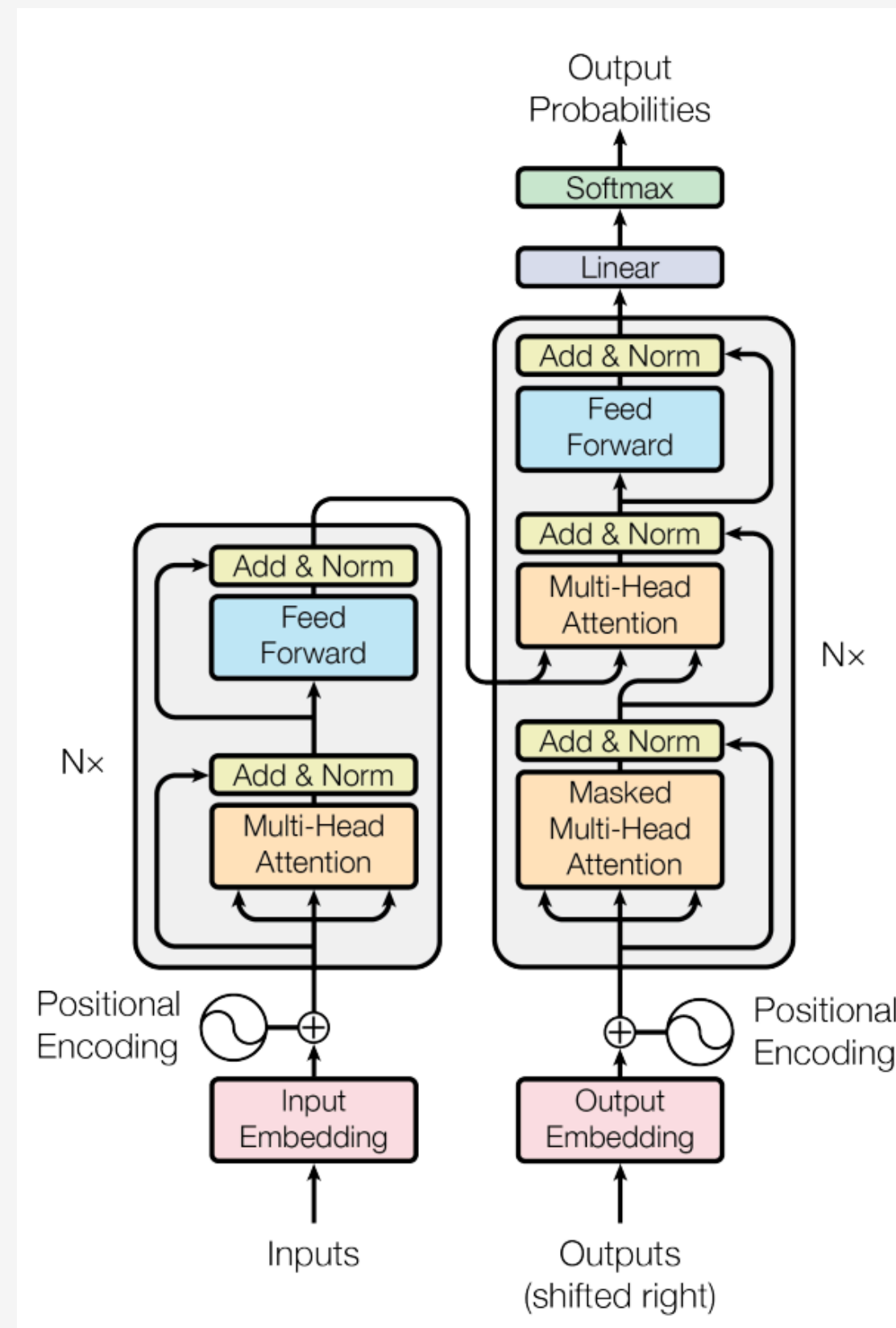


[2] 20230627_GenAI_on_AWS_edutech_forum_sungeuns

History of GPT Series: Transformer

Encoder

- 주어진 텍스트를 이해하기 위한 component
 - 즉, 언어 이해 담당
- > Google의 BERT 시리즈



[3]

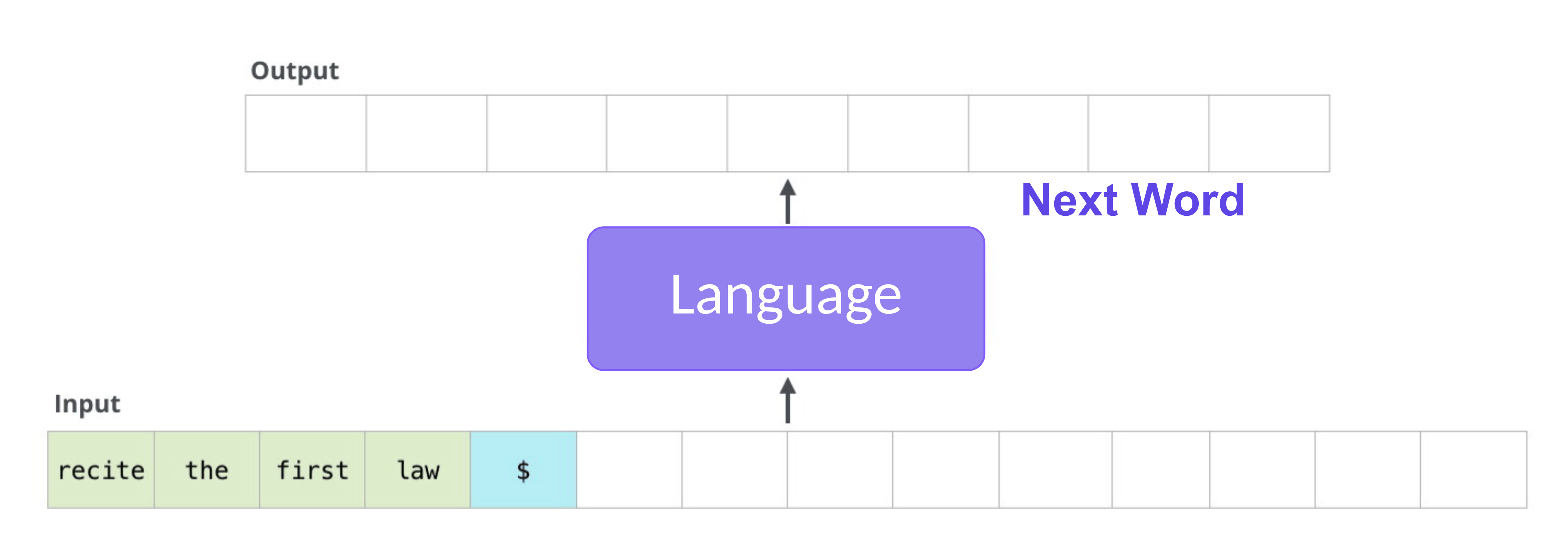
Decoder

- 이해한 텍스트를 기반으로 언어를 생성해내는 component
 - 즉, 언어 생성 담당
- > OpenAI의 GPT 시리즈

[3] Vaswani, Ashish, et al. "Attention is all you need."

History of GPT Series: How to train?

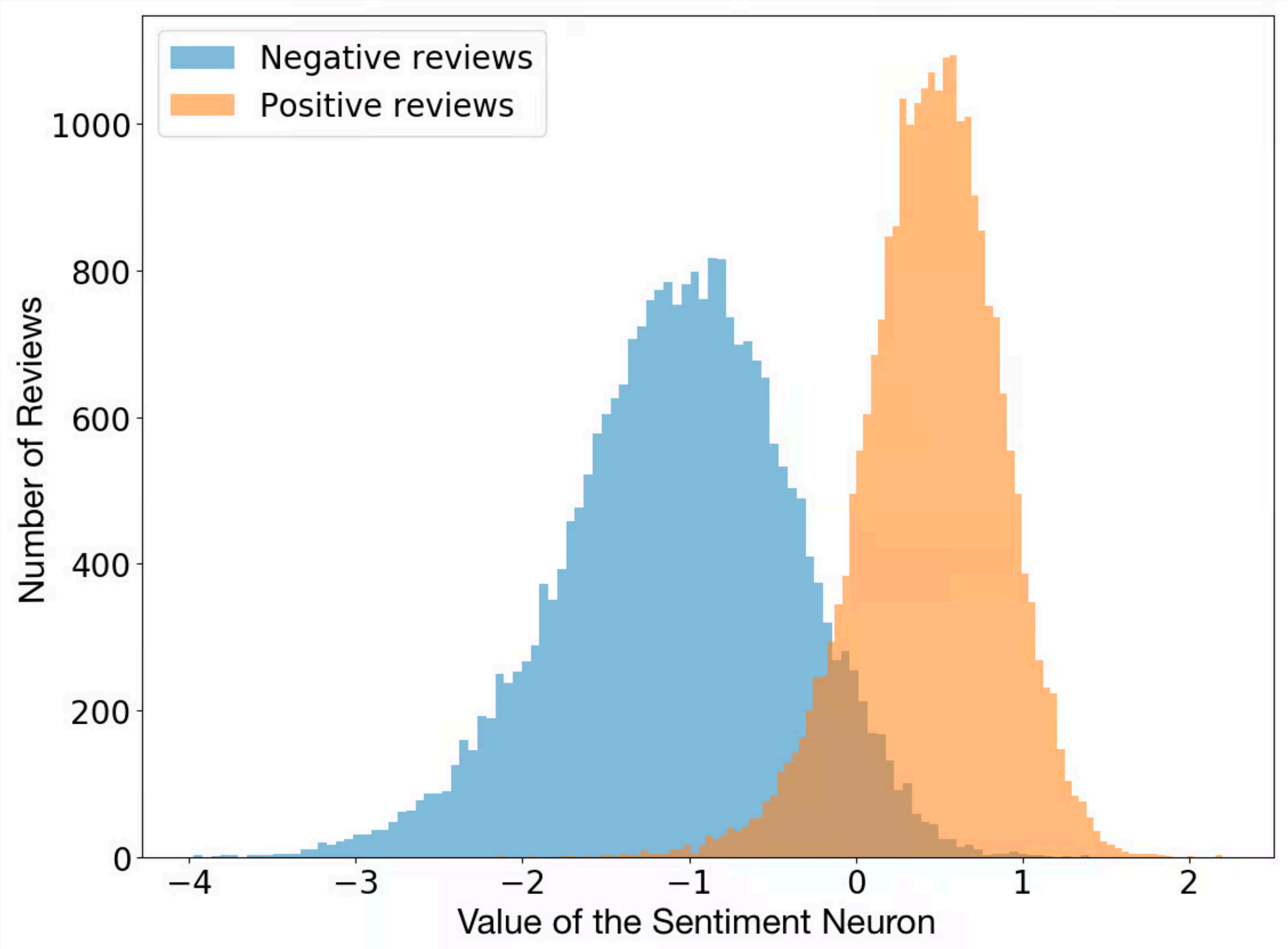
[4]



[4] https://ratsgo.github.io/nlpbook/docs/language_model/bert_gpt/

History of GPT Series: Emergent Ability

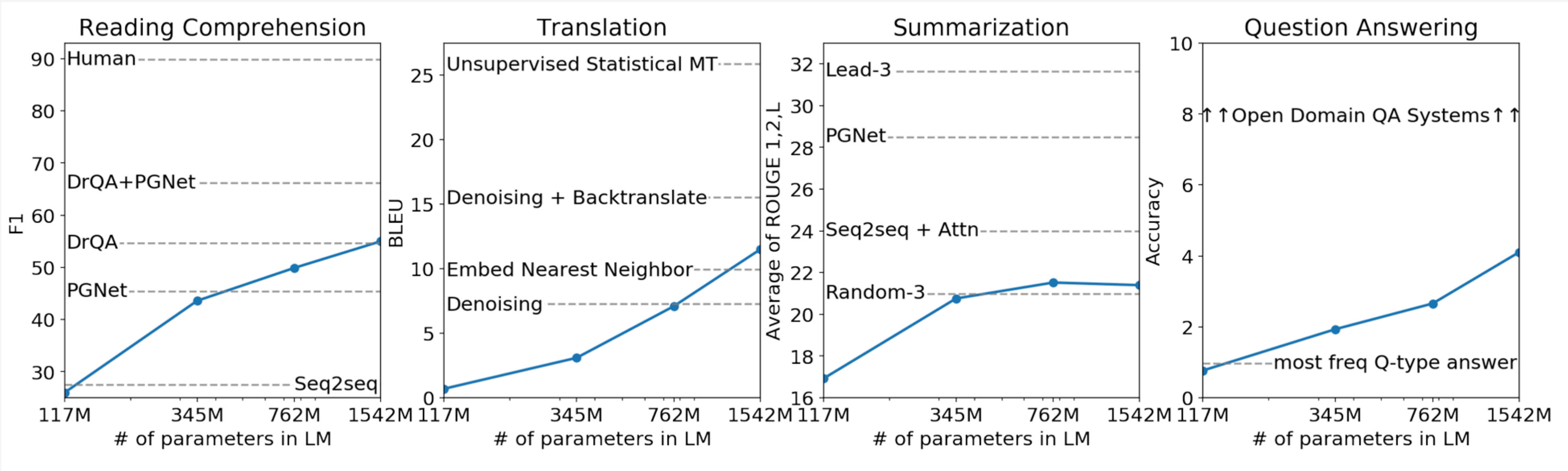
[5]



[5] <https://openai.com/research/unsupervised-sentiment-neuron>

History of GPT Series: GPT-2

Zero-shot Learning

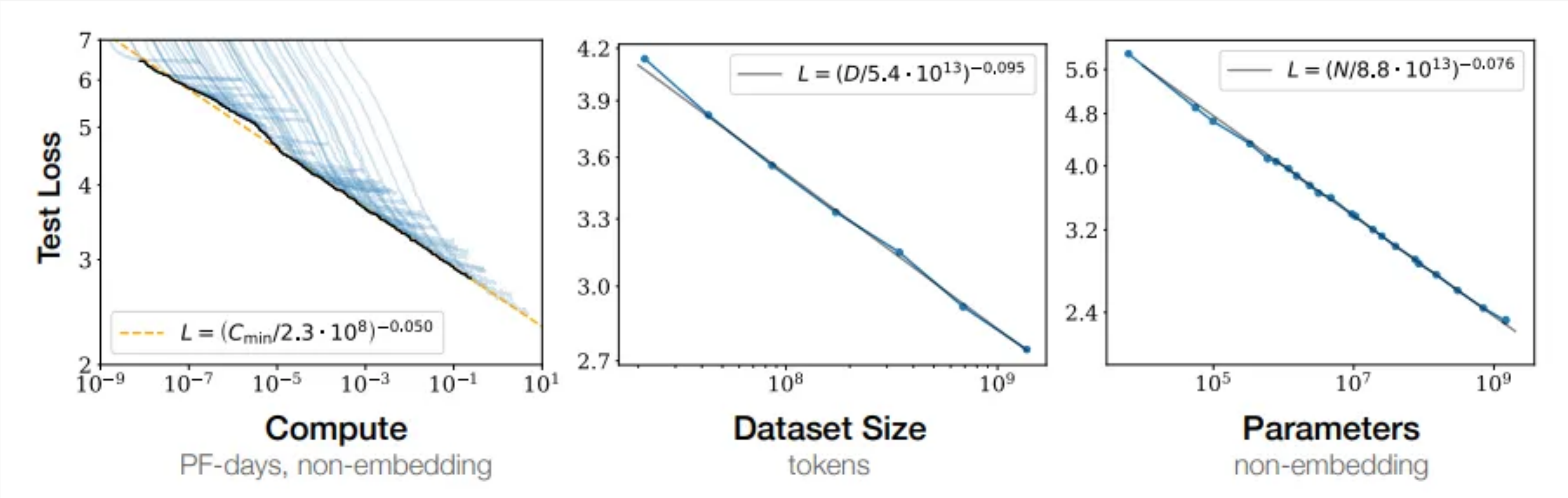


[6]

[6] Radford, Alec, et al. "Language models are unsupervised multitask learners."

History of GPT Series: Scaling Law

[7]

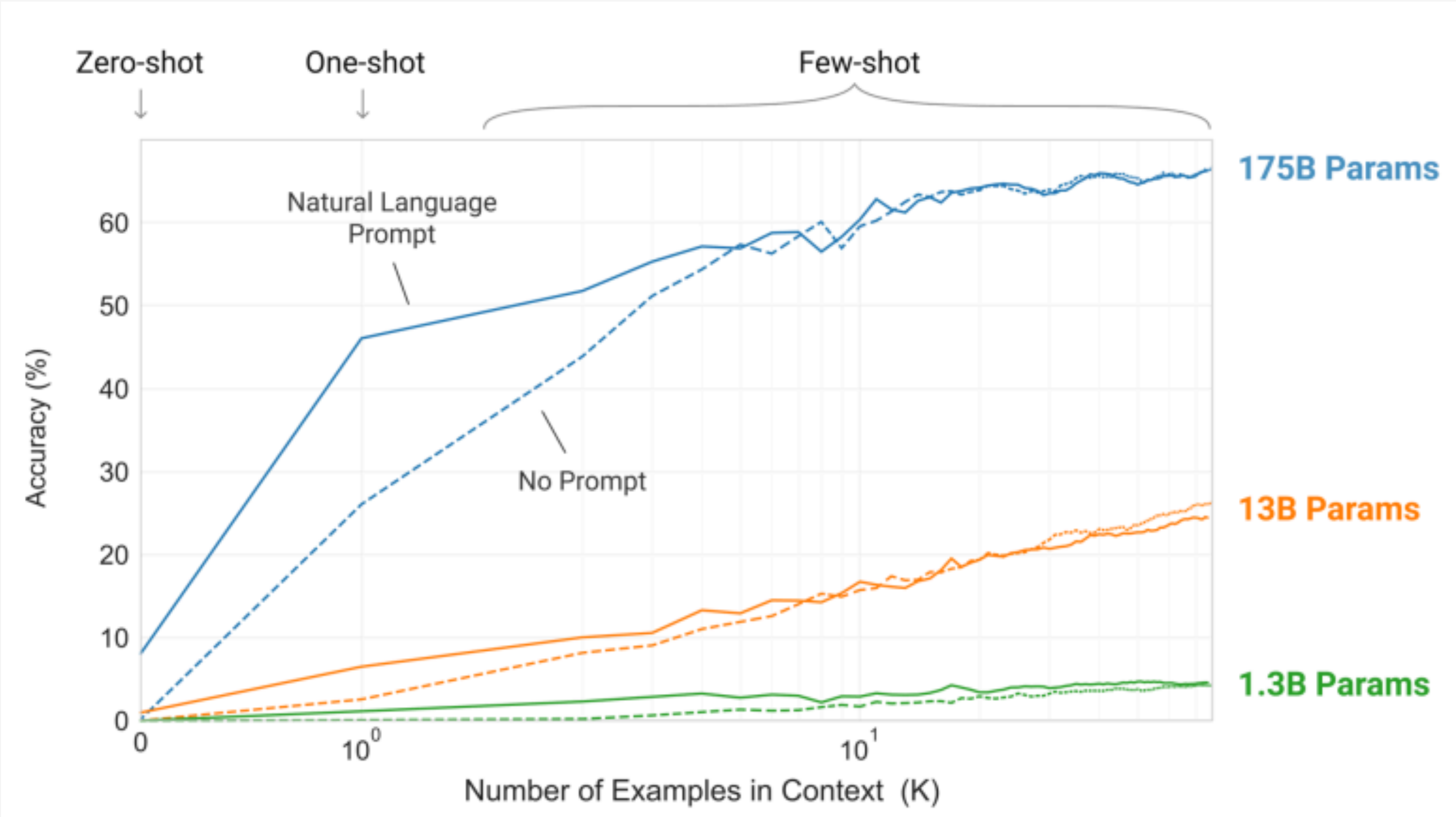


[7] Kaplan, Jared, et al. "Scaling laws for neural language models."

History of GPT Series: GPT-3

In-context Learning

[8]



[8] Brown, Tom, et al. "Language models are few-shot learners."

History of GPT Series: InstructGPT

Steerability

[9]

Prompt *Explain the moon landing to a 6 year old in a few sentences.*

Completion

GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

[9] <https://openai.com/research/instruction-following>

History of GPT Series: Instruction Tuning for Steerability

[10]

Supervised Learning

Instruction: Given an address and city, come up with the zip code.

Input:

Address: 123 Main Street, City: San Francisco

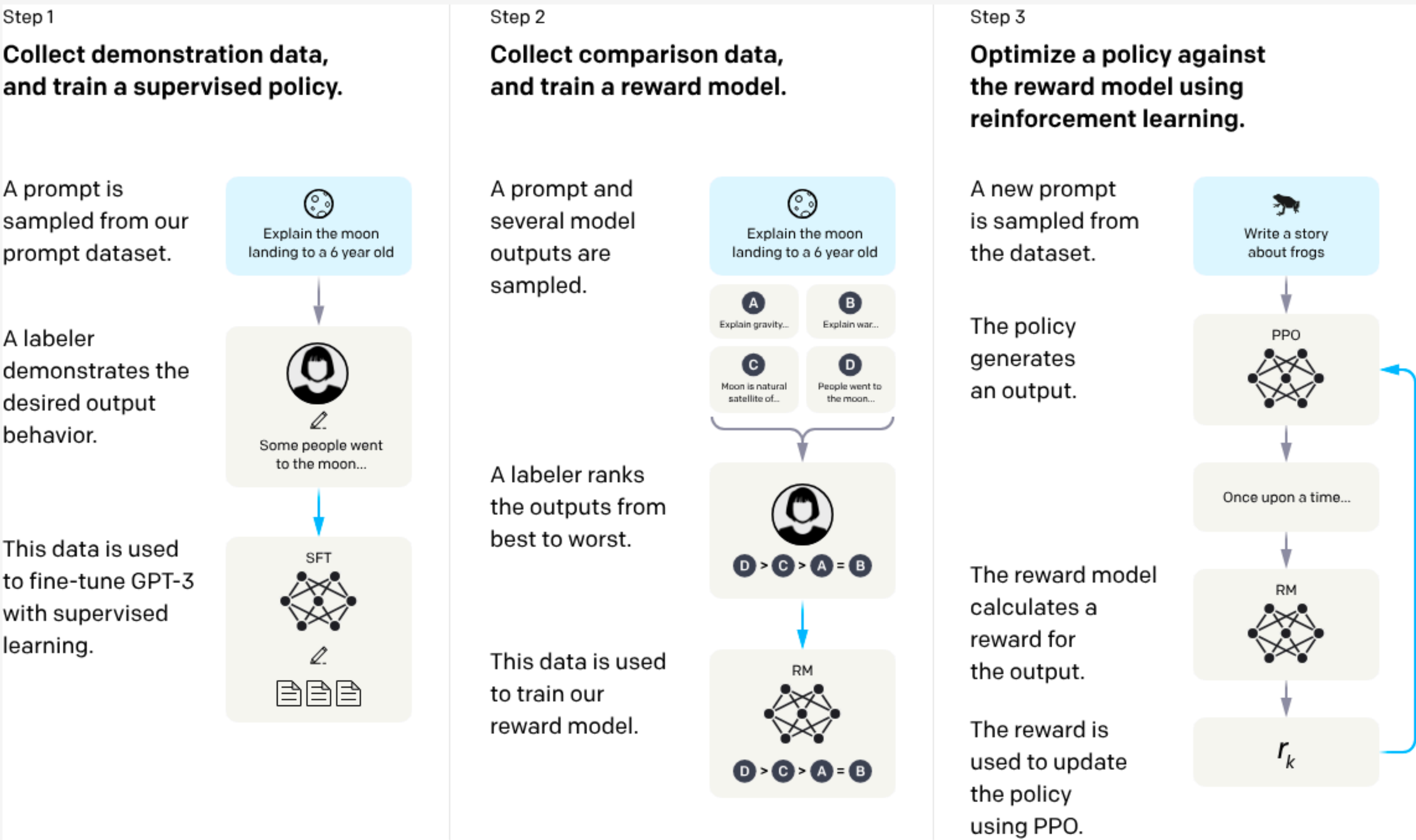
Output: 94105



[10] Wang, Yizhong, et al. "Self-instruct: Aligning language model with self generated instructions."

History of GPT Series: Alignment Tuning for Steerability

Reinforcement Learning from Human Feedback (RLHF)



[9] <https://openai.com/research/instruction-following>

How to improve LLMs ?

Prompt Engineering

Prompt

Context

Example

Instruction

System:
You are an AI assistant that helps people find information. User will give you a question. Your task is to answer as faithfully as you can. While answering think step-by-step and justify your answer.

Human:
John had 10 less apples than 20. Which represents the number of apples John had?
(A) 5
(B) 9
(C) 11
(D) 20
(E) None of the above

Assistant:
Let's represent the number of apples John had as x. According to the problem, John had 10 less apples than 20, so we can write the equation:
 $x = 20 - 10$
Now, solve for x:
 $x = 10$
So, the number of apples John had is 10. This is not in the given options, so the correct answer is:
(E) None of the above

Human:
Pick which sentence is not logical.
Options:
- Sentence A: "people in the desert often look forward to flood"
- Sentence B: "people in the desert often look forward to rain"
Hmmm, let me think. I want to lay out the solution in details.

[11]

[11] Mukherjee, Subhabrata, et al. "Orca: Progressive learning from complex explanation traces of gpt-4."

Prompting Strategies: (1) Clarity with Delimiters

Unclear Prompt:

The description for this product should be fairly short, a few sentences only, and not too much more.

Clear Prompt:

Use a 3 to 5 sentence paragraph to describe this product.

Less Effective Prompt:

Summarize the text below as a bullet point list of the most important points.

{text input here}

Better Prompt:

Summarize the text below as a bullet point list of the most important points.

Text: ""

{text input here}

""

Prompting Strategies: (2) Few-shot Examples

System:

You are an AI assistant that helps people find information. User will give you a question. Your task is to answer as faithfully as you can. While answering think step-by-step and justify your answer.

Human:

Pick which sentence is not logical.

Options:

- Sentence A: "people in the desert often look forward to flood"
- Sentence B: "people in the desert often look forward to rain"

Hmmm, let me think. I want to lay out the solution in details.

System:

You are an AI assistant that helps people find information. User will give you a question. Your task is to answer as faithfully as you can. While answering think step-by-step and justify your answer.

Human:

John had 10 less apples than 20. Which represents the number of apples John had?

- (A) 5
- (B) 9
- (C) 11
- (D) 20
- (E) None of the above

Assistant:

Let's represent the number of apples John had as x . According to the problem, John had 10 less apples than 20, so we can write the equation:

$$x = 20 - 10$$

Now, solve for x :

$$x = 10$$

So, the number of apples John had is 10. This is not in the given options, so the correct answer is:

- (E) None of the above

Human:

Pick which sentence is not logical.

Options:

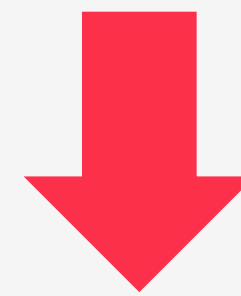
- Sentence A: "people in the desert often look forward to flood"
- Sentence B: "people in the desert often look forward to rain"

Hmmm, let me think. I want to lay out the solution in details.

Prompting Strategies: (3) Understanding about LLMs

 **You**
지금 시애틀 시간은 몇 시야?

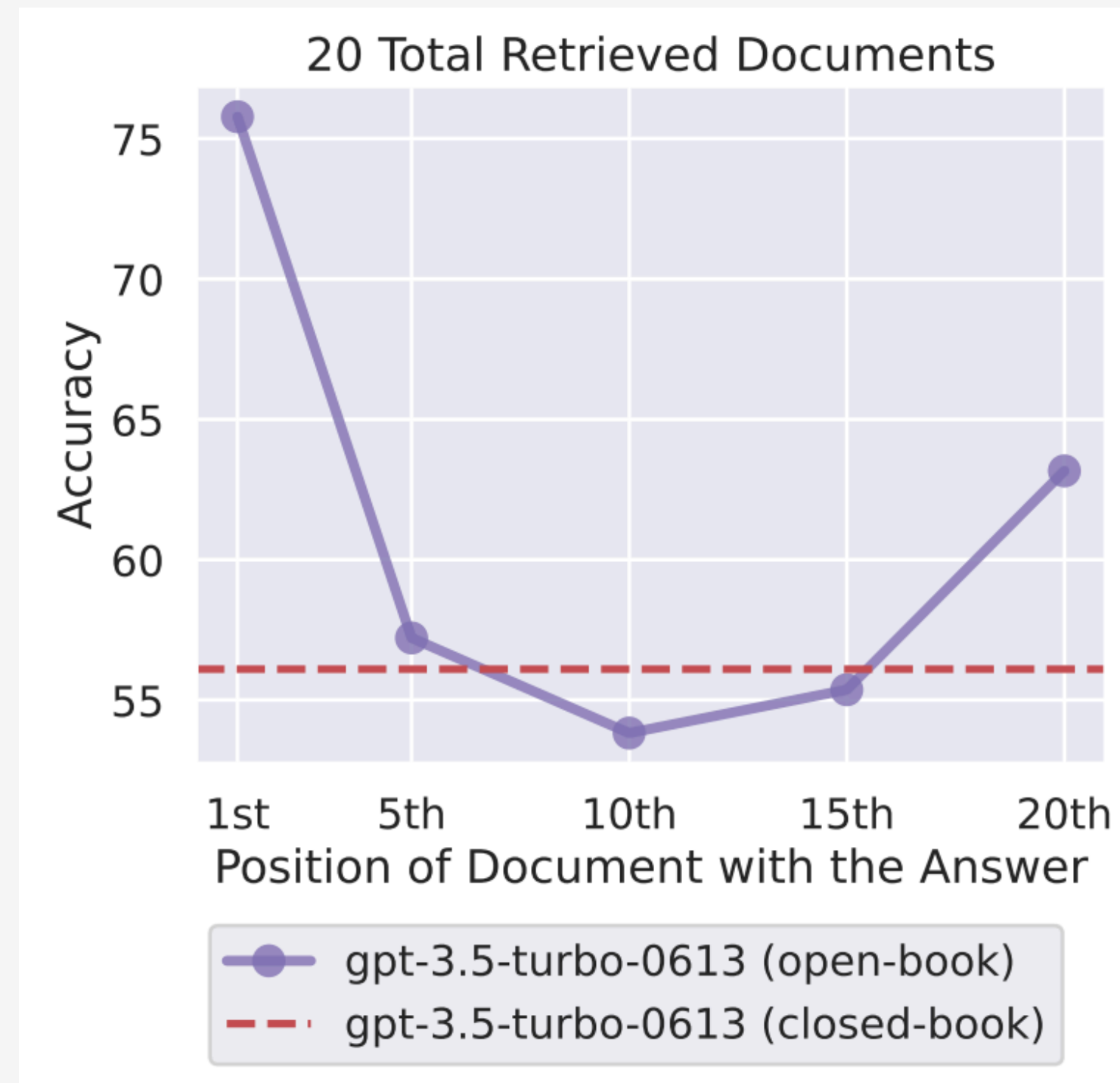
 **ChatGPT**
죄송합니다. 시애틀의 현재 시간은 알려드릴 수 없습니다. 하지만 인터넷 또는 시계에서 쉽게 확인할 수 있습니다.



 **You**
한국이 지금 오후 6시 14분인데, 시애틀 시간은 몇 시야?

 **ChatGPT**
한국의 현재 시간이 오후 6시 14분이라면, 시애틀은 태평양 시간대(Pacific Time Zone)에 있으므로 시애틀 시간은 오전 1시 14분입니다. 태평양 시간대는 한국 시간보다 17시간 뒤에 있습니다.

Prompting Strategies: (4) Position of Information



[12] Liu, Nelson F., et al. "Lost in the middle: How language models use long contexts."

Prompting Strategies: (5) Chain-of-Thought (CoT)

[13]

Standard Prompting	Chain-of-Thought Prompting
Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: The answer is 11. Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?	Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11. Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?
Model Output A: The answer is 27. ❌	Model Output A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

[13] Wei, Jason, et al. "Chain-of-thought prompting elicits reasoning in large language models."

Prompting Strategies: (6) Program-of-Thought (PoT)

[14]

Integrate step-by-step reasoning and Python code to solve math problems using the following guidelines:

- Analyze the question and write functions to solve the problem; the function should not take any arguments.
- Present the final result in LaTeX using a `\boxed{ }` without any units.
- Utilize the `'pi'` symbol and `'Rational'` from Sympy for π and fractions, and simplify all fractions and square roots without converting them to decimal values.

Question: Compute $1011_2 + 101_2 - 1100_2 + 1101_2$. Express your answer in base 2.

Solution: Let's write a single python function for binary computation:

```
“python
def binary_sum_diff():
    # Convert binary to decimal
    num1 = int("1011", 2)
    num2 = int("101", 2)
    num3 = int("1100", 2)
    num4 = int("1101", 2)
    # Perform the operations
    result = num1 + num2 - num3 + num4
    # Convert the result back to binary
    result_binary = format(result, "b")
    return result_binary
result = binary_sum_diff()
print(result)
”

“output
10001
”

The answer expressed in base 2 is \boxed{10001_2}.
```

[14] Gou, Zhibin, et al. "Tora: A tool-integrated reasoning agent for mathematical problem solving."

Prompting Strategies: (7) Others...

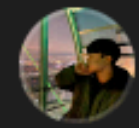
[15]

Source	Instruction	Acc
<i>Baselines</i>		
(Kojima et al., 2022)	Let's think step by step.	71.8
(Zhou et al., 2022b)	Let's work this out in a step by step way to be sure we have the right answer.	58.8
	(empty string)	34.0
<i>Ours</i>		
PaLM 2-L-IT	Take a deep breath and work on this problem step-by-step.	80.2
PaLM 2-L	Break this down.	79.9
gpt-3.5-turbo	A little bit of arithmetic and a logical approach will help us quickly arrive at the solution to this problem.	78.5
gpt-4	Let's combine our numerical command and clear thinking to quickly and accurately decipher the answer.	74.5

[15] Yang, Chengrun, et al. "Large language models as optimizers."

Retrieval-Augmented Generation (RAG)

Limitation of LLMs: Lack of Recent Information



You

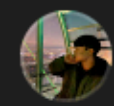
2023 카타르 아시안컵의 우승국은 어디야 ?



ChatGPT

죄송하지만, 제가 2023년 아시안컵의 결과를 알 수는 없습니다. 제가 마지막으로 업데이트된 정보는 2022년 1월 까지이며, 그 이후의 사건에 대한 정보는 제게 알려지지 않았습니니다. 아시안컵의 우승자에 대한 정보를 확인하려면 인터넷 뉴스나 스포츠 관련 웹사이트를 참조하시기 바랍니다.

Limitation of LLMs: Hallucination



You

What is the Upstage ?



ChatGPT

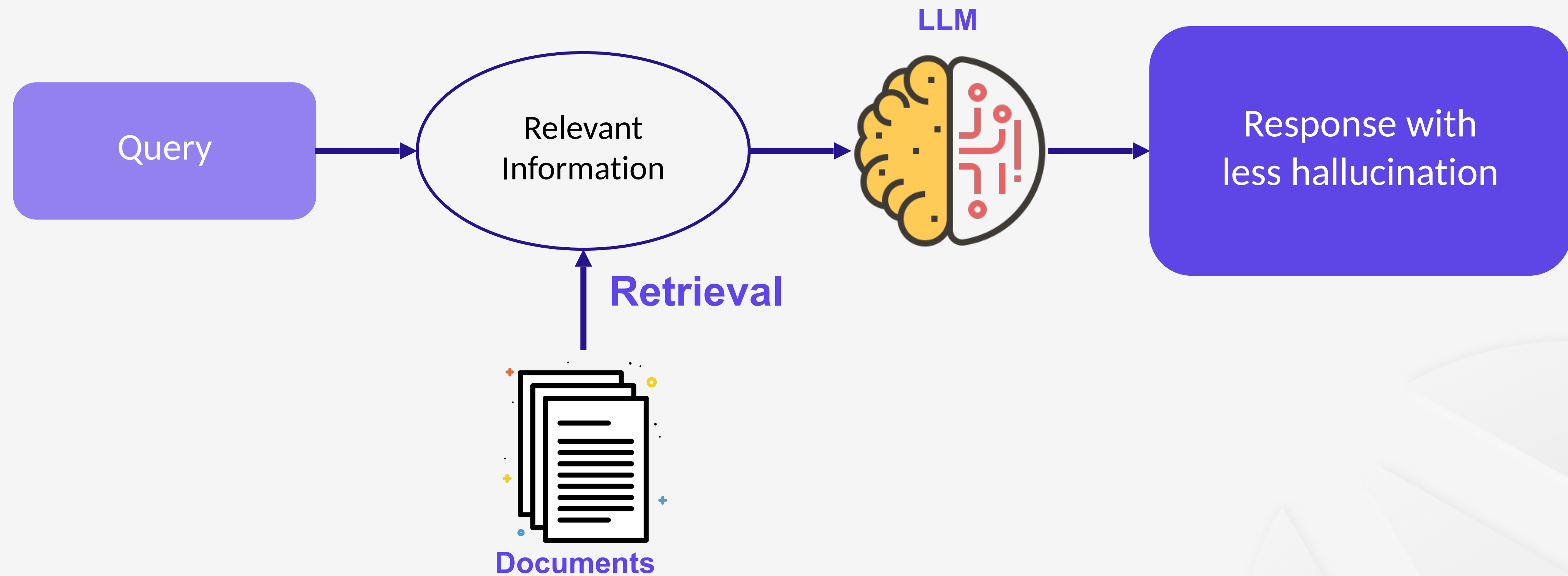
"Upstage" can refer to several different things depending on the context:

1. **Theater Terminology:** In theater, "upstage" refers to the part of the stage that is farthest from the audience. Conversely, "downstage" is the part closest to the audience. These terms are derived from the days when stages were raked (sloped), and actors had to literally walk uphill (upstage) or downhill (downstage) to move closer or farther from the audience.
2. **Upstaging (verb):** This refers to when one actor deliberately or inadvertently draws attention to themselves, diverting attention from another actor who is supposed to be the focus of the scene.
3. **Upstage (noun):** There's also a concept of "upstaging" in the sense of social media or online presence, where someone or something might "upstage" another by drawing more attention or generating more interest.

Could you please clarify which meaning you're referring to, or if it's something else entirely?

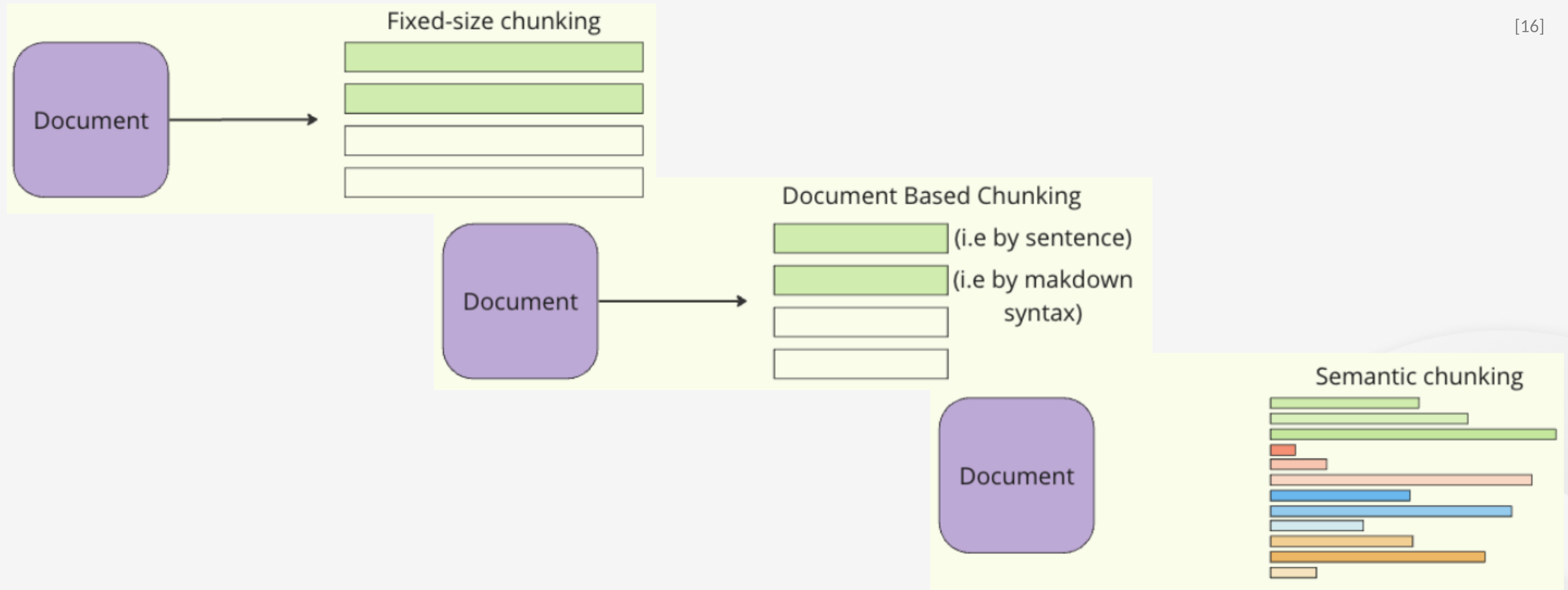
Overview of RAG

[16]



[16] <https://miro.com/app/board/uXjVNvklNmc=/>

How to Improve RAG ? : Chunking and Embedding

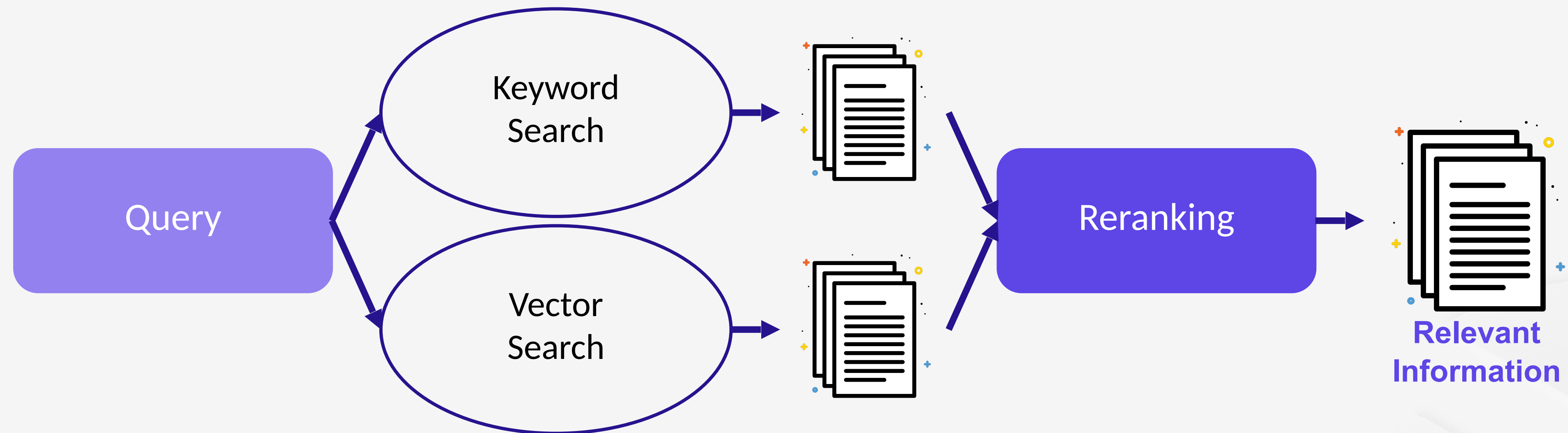


[16] <https://miro.com/app/board/uXjVNvklNmc=/>

How to Improve RAG?: Retrieval Strategy (1)

Hybrid Search

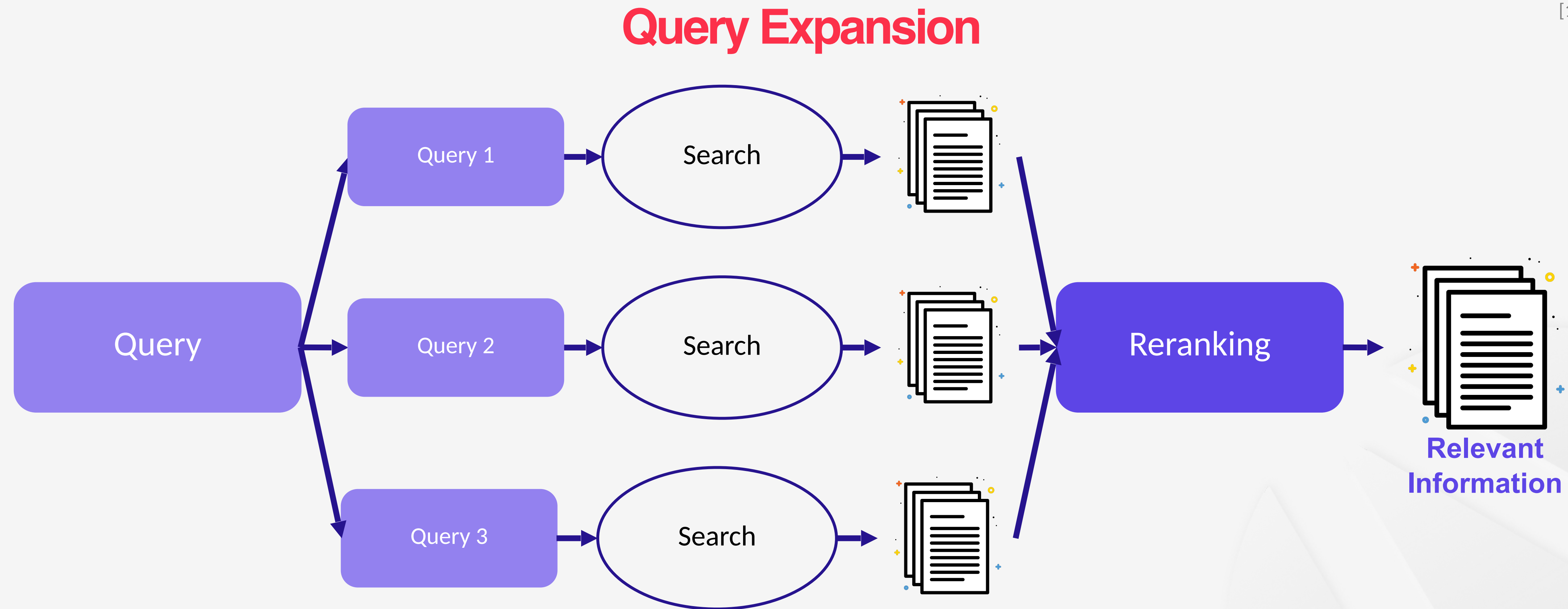
[16]



[16] <https://miro.com/app/board/uXjVNvklNmc=/>

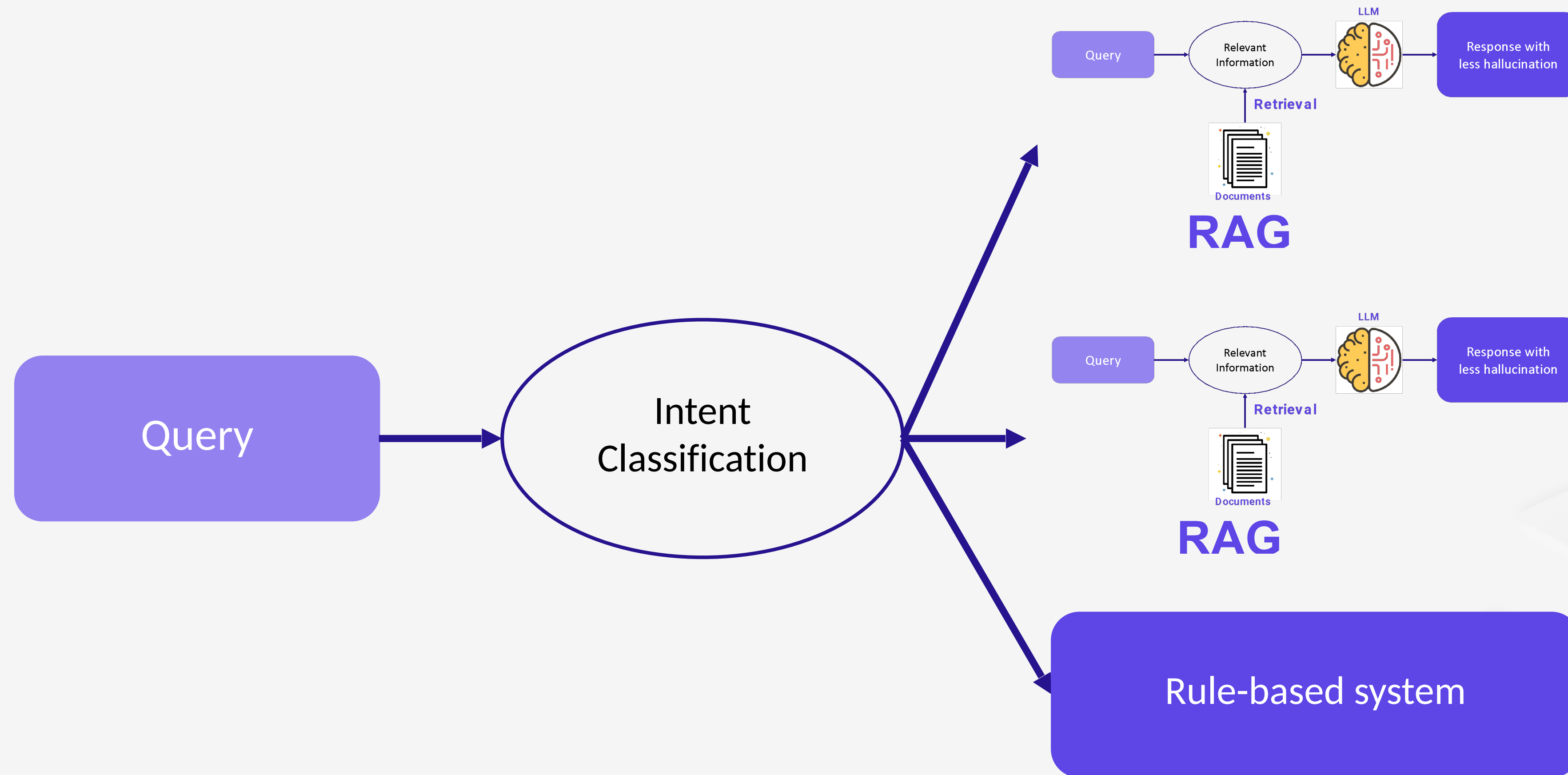
How to Improve RAG ? : Retrieval Strategy (2)

[16]



[16] <https://miro.com/app/board/uXjVNvklNmc=/>

How to Improve RAG ? : Retrieval Strategy (3)

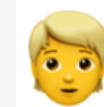


Instruction Tuning

What is Instruction Tuning ?

Supervised Learning based on Only Response Loss

[17]



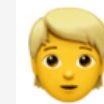
<|user|>

Explain the fault-tolerance of the reaction control system on the Space Shuttle.



<|assistant|>

The reaction control system (RCS) on the Space Shuttle was designed to be fault-tolerant, meaning it was able to continue functioning even if one or more of its components failed. The RCS consisted of two sets of ... </s>



<|user|>

Did the RCS have any on-orbit failures?



<|assistant|>

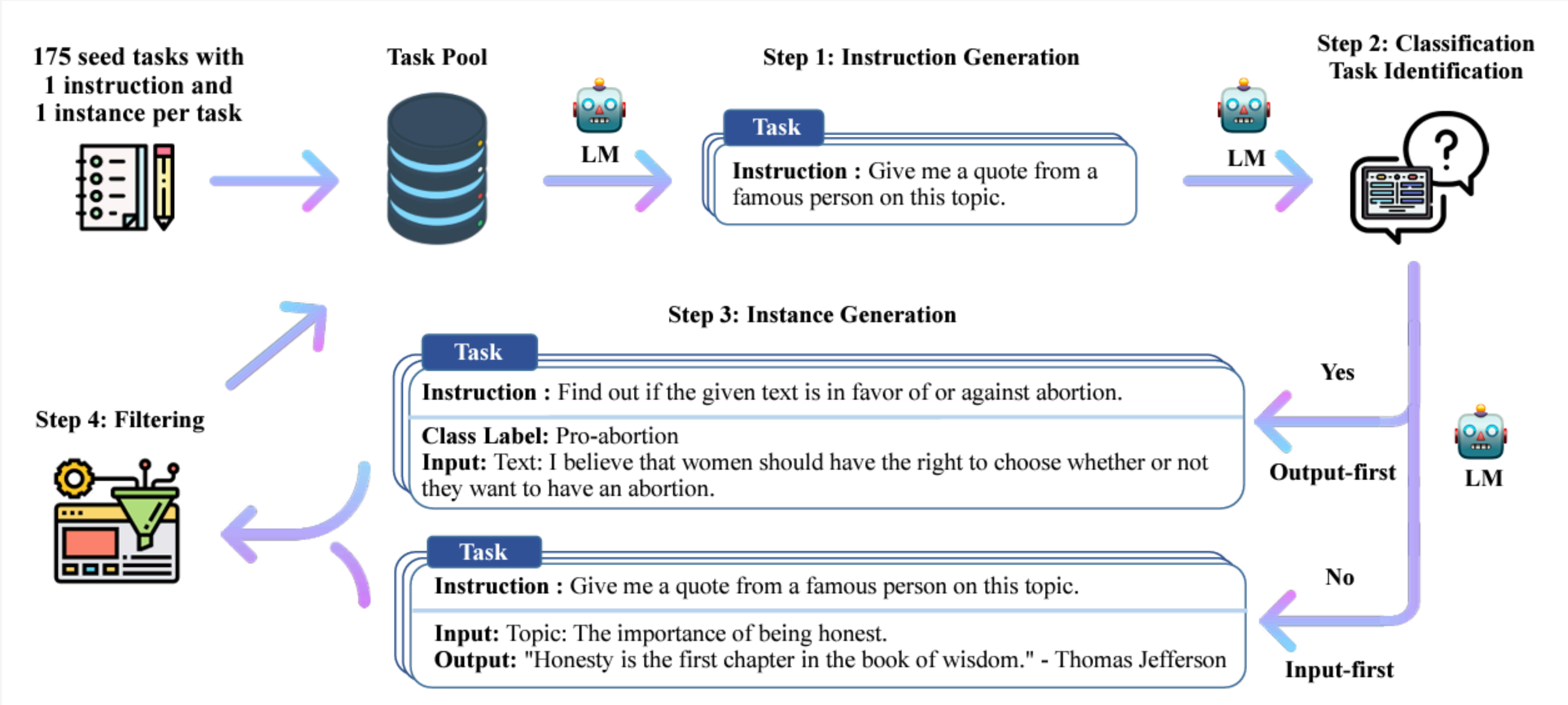
There were several instances where the reaction control system (RCS) on the Space Shuttle experienced failures or malfunctions during on-orbit missions. These ... </s>

[17] Wang, Yizhong, et al. "How far can camels go? exploring the state of instruction tuning on open resources."

How to Make Instruction Dataset?: Synthetic Manner (1)

Self-instruct

[18]



[18] Wang, Yizhong, et al. "Self-instruct: Aligning language model with self generated instructions."

How to Make Instruction Dataset?: Synthetic Manner (2)

Orca

[19]

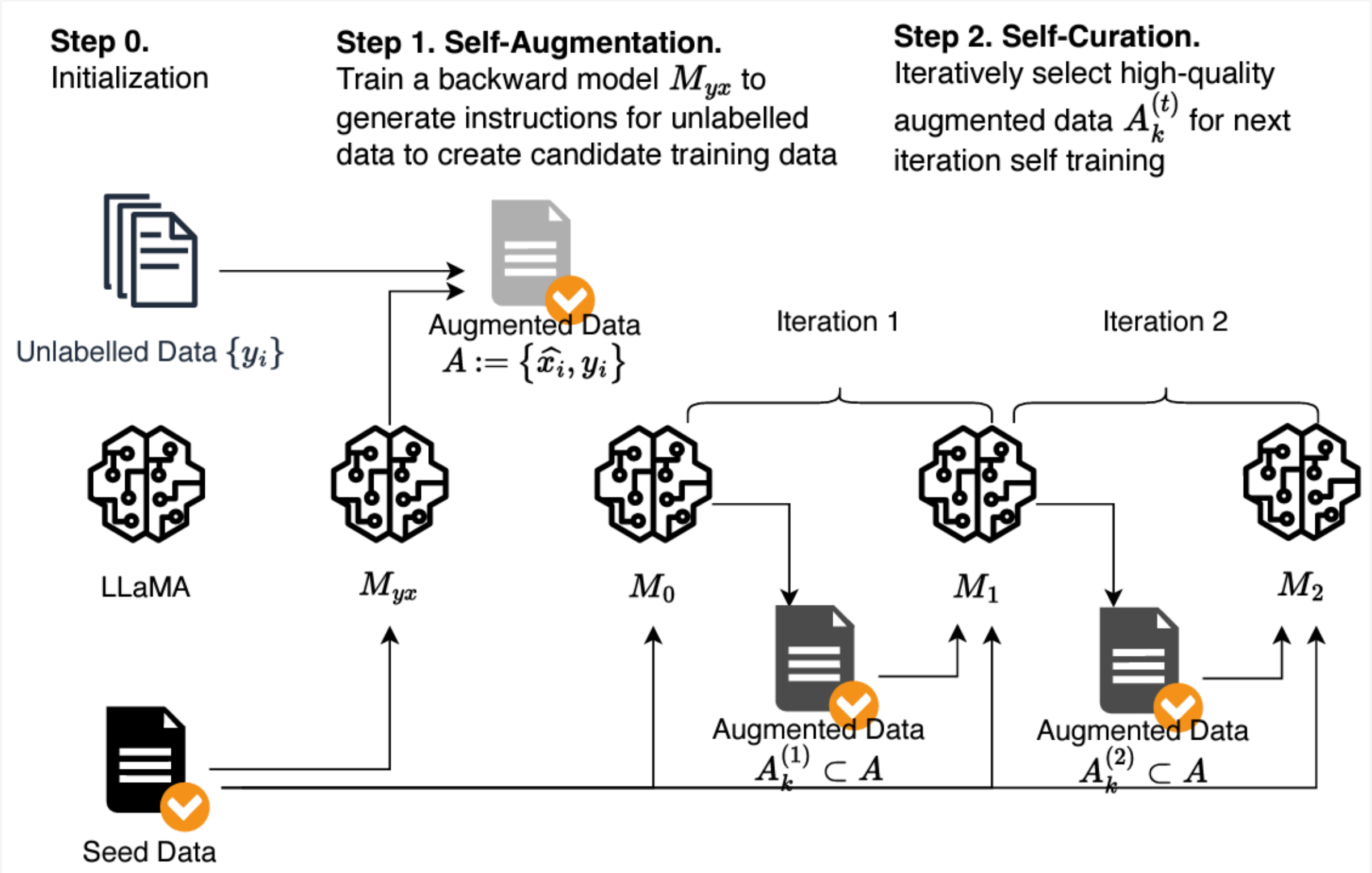
Id.	System Message
1	<empty system message>
2	You are an AI assistant. Provide a detailed answer so user don't need to search outside to understand the answer.
3	You are an AI assistant. You will be given a task. You must generate a detailed and long answer.
4	You are a helpful assistant, who always provide explanation. Think like you are answering to a five year old.
5	You are an AI assistant that follows instruction extremely well. Help as much as you can.
6	You are an AI assistant that helps people find information. Provide a detailed answer so user don't need to search outside to understand the answer.
7	You are an AI assistant. User will you give you a task. Your goal is to complete the task as faithfully as you can. While performing the task think step-by-step and justify your steps.
8	You should describe the task and explain your answer. While answering a multiple choice question, first output the correct answer(s). Then explain why other answers are wrong. Think like you are answering to a five year old.
9	Explain how you used the definition to come up with the answer.
10	You are an AI assistant. You should describe the task and explain your answer. While answering a multiple choice question, first output the correct answer(s). Then explain why other answers are wrong. You might need to use additional knowledge to answer the question.
11	You are an AI assistant that helps people find information. User will you give you a question. Your task is to answer as faithfully as you can. While answering think step-by-step and justify your answer.
12	User will you give you a task with some instruction. Your job is follow the instructions as faithfully as you can. While answering think step-by-step and justify your answer.
13	You are a teacher. Given a task, you explain in simple steps what the task is asking, any guidelines it provides and how to use those guidelines to find the answer.
14	You are an AI assistant, who knows every language and how to translate one language to another. Given a task, you explain in simple steps what the task is asking, any guidelines that it provides. You solve the task and show how you used the guidelines to solve the task.
15	Given a definition of a task and a sample input, break the definition into small parts. Each of those parts will have some instruction. Explain their meaning by showing an example that meets the criteria in the instruction. Use the following format: Part #: a key part of the definition. Usage: Sample response that meets the criteria from the key part. Explain why you think it meets the criteria.
16	You are an AI assistant that helps people find information.

[19] Mukherjee, Subhabrata, et al. "Orca: Progressive learning from complex explanation traces of gpt-4."

How to Make Instruction Dataset?: Synthetic Manner (3)

Self-alignment

[20]



[20] Li, Xian, et al. "Self-alignment with instruction backtranslation."

How to Make Instruction Dataset?: Synthetic Manner (4)

SAIL

[21]

(a) Standard Prompt

Below is an instruction that describes a task.
Write a response that appropriately completes the request.

Instruction: [Instruction]
Input: [Input or None]

Response:

(b) Search-Augmented Prompt

Below is an instruction that describes a task.
Write a response that appropriately completes the request.

Related Information:
[Title 3]\n [Preview 3]
[Title 2]\n [Preview 2]
[Title 1]\n [Preview 1]

Instruction: [Instruction]
Input: [Input or None]

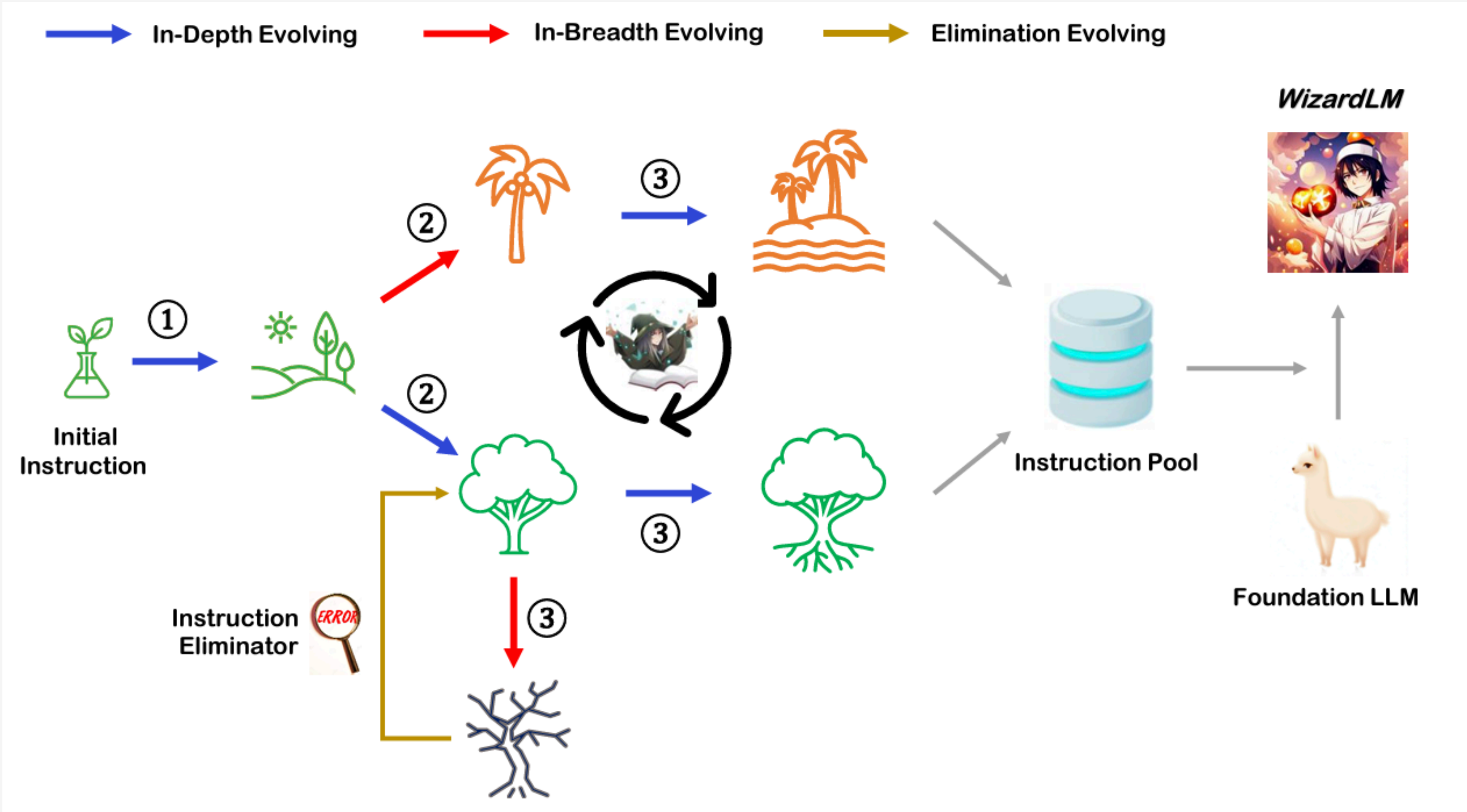
Response:

[21] Luo, Hongyin, et al. "SAIL: Search-Augmented Instruction Learning."

How to Make Instruction Dataset?: Synthetic Manner (5)

WizardLM

[22]



[22] Xu, Can, et al. "Wizardlm: Empowering large language models to follow complex instructions."

Alignment Tuning

Why Alignment Tuning ?



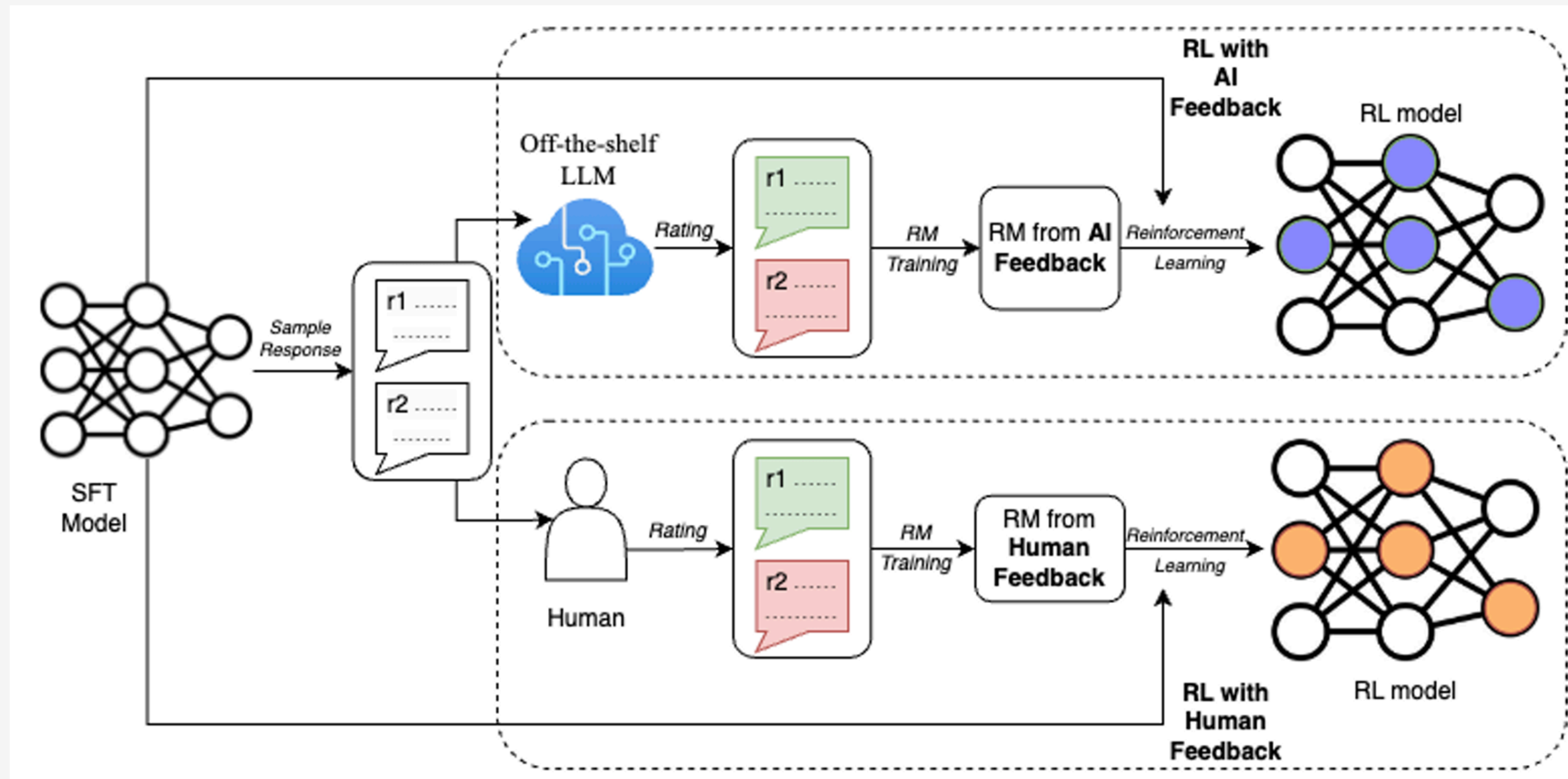
Instruction tuning for alignment

- Not synthetic instruction dataset, but human expert-written instruction dataset -> High cost !!
- However, relative human judgments of response quality are often much easier!

Alignment Tuning based on Reinforcement Learning

RLHF and RLAIIF

[23]



[23] Lee, Harrison, et al. "Rlaif: Scaling reinforcement learning from human feedback with ai feedback."

Limitations of Alignment Tuning based on RL

Point 1

Unstable

- RL 기반 PPO가 SFT 대비 unstable하다고 알려져 있음

Point 2

Too complex

- PPO는 수많은 hyper-params.를 필요로 함
- 그러한 hyper-params.에 sensitive 함

Point 3

Memory burden

- PPO를 위해서는, 동시에 많은 models을 memory에 올려야 함
- Policy model, value model, reward model, reference model

Alignment Tuning based on SFT (1)

RRHF

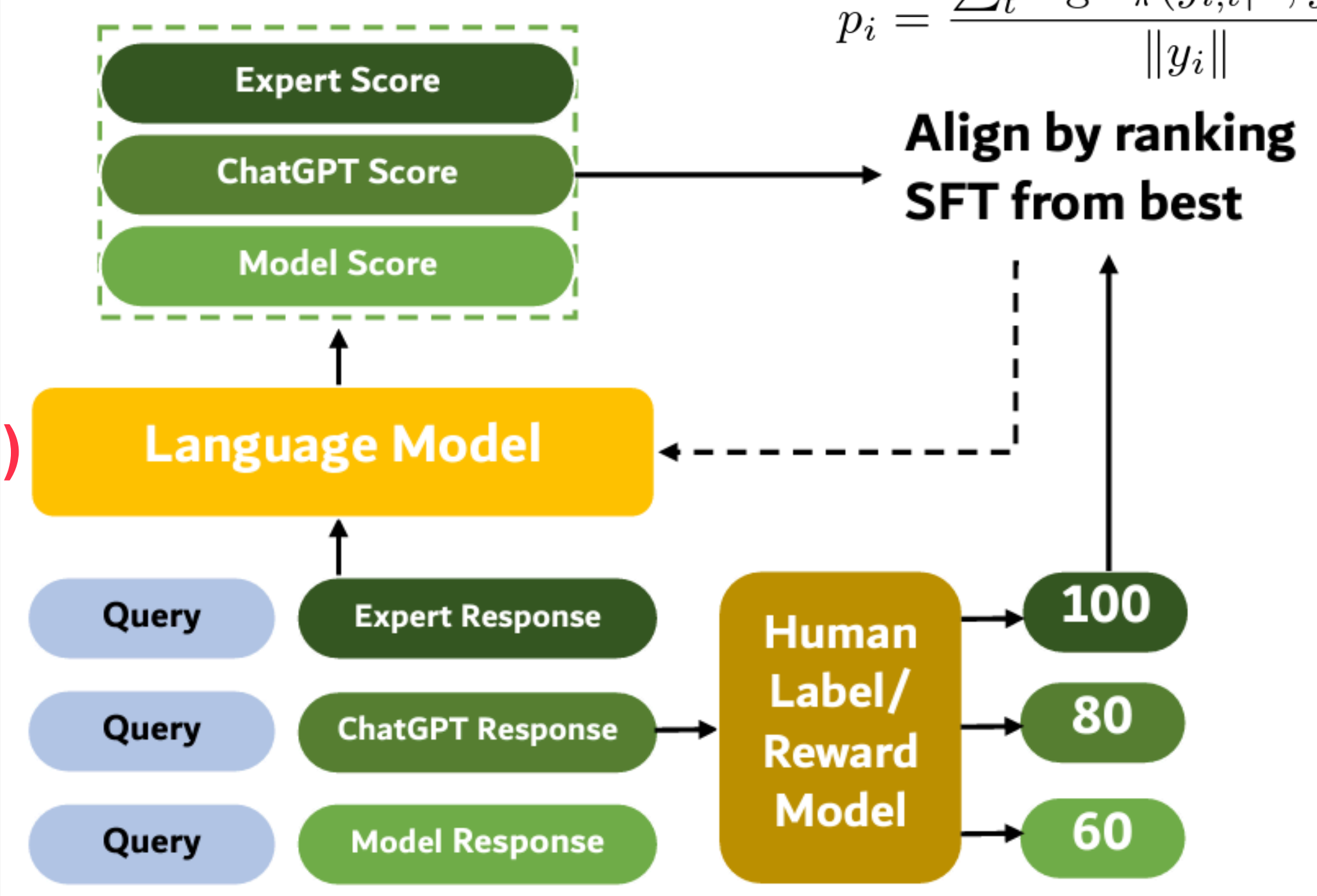
[24]

Conditional log prob.

$$p_i = \frac{\sum_t \log P_{\pi}(y_{i,t}|x, y_{i,<t})}{\|y_i\|}$$

$$L_{rank} = \sum_{r_i < r_j} \max(0, p_i - p_j)$$

Reward model (=)



[24] Yuan, Zheng, et al. "Rrhf: Rank responses to align language models with human feedback without tears."

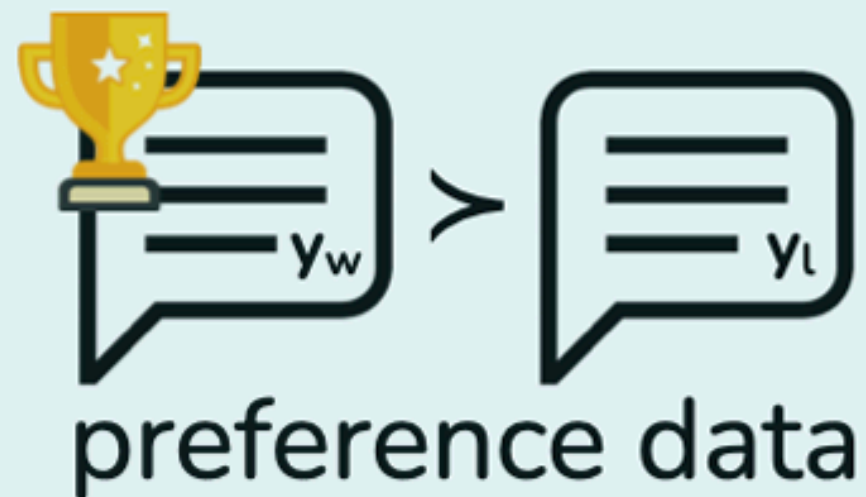
Alignment Tuning based on SFT (2)

DPO

[25]

Direct Preference Optimization (DPO)

x : "write me a poem about
the history of jazz"



maximum
likelihood

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

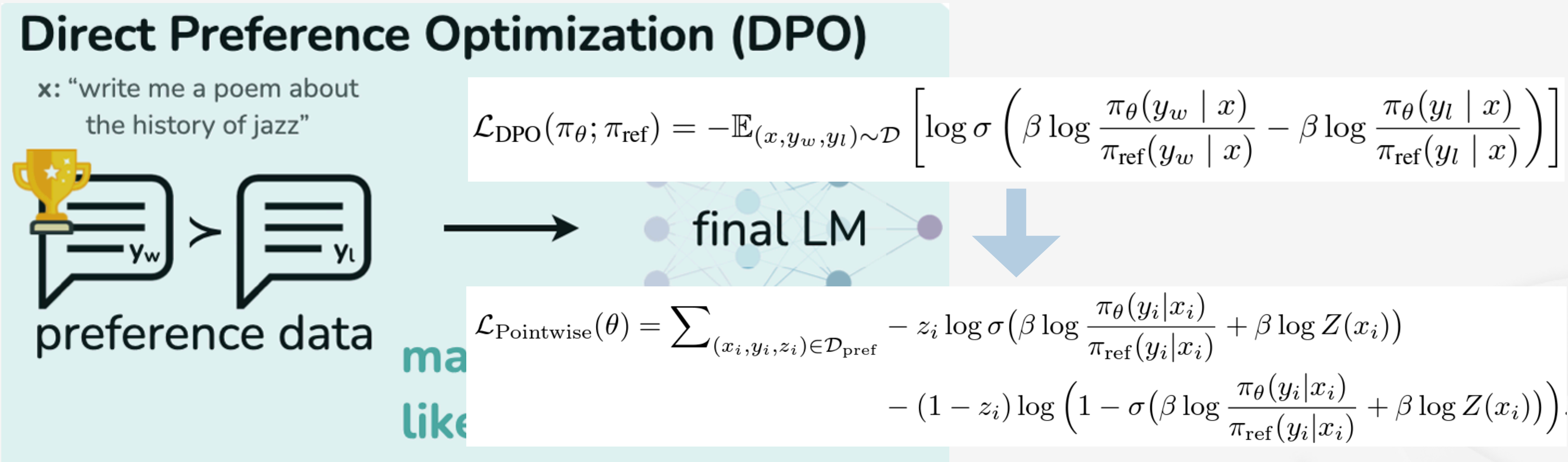


[25] Rafailov, Rafael, et al. "Direct preference optimization: Your language model is secretly a reward model."

Alignment Tuning based on SFT (3)

ULMA (point-wise DPO)

[26]

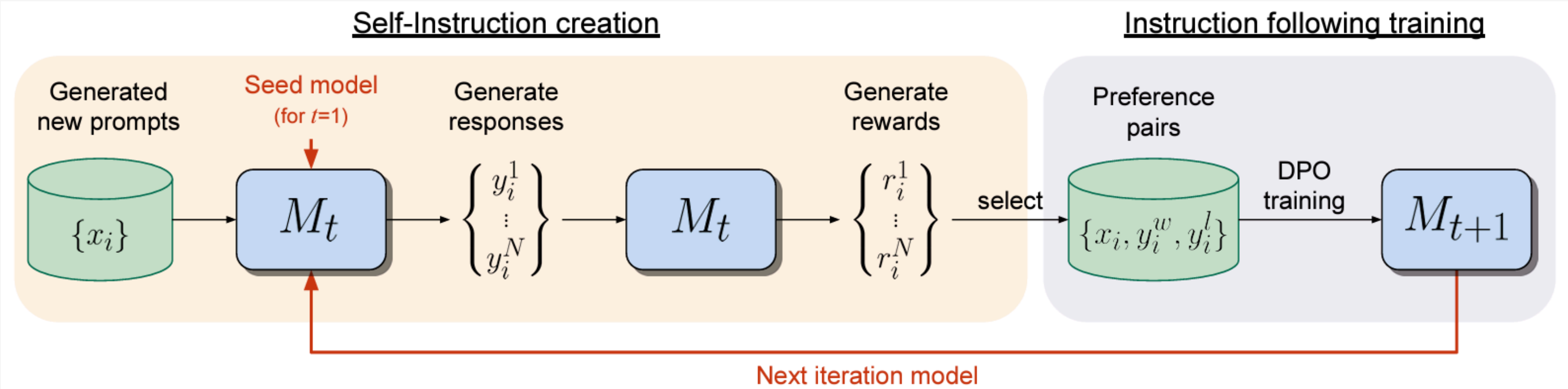


[26] Cai, Tianchi, et al. "ULMA: Unified Language Model Alignment with Demonstration and Point-wise Human Preference."

Alignment Tuning based on SFT (4)

Self-rewarding

[27]



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Thank you :)